Forecasting IT Sector Growth in Bangladesh: A Data Science-Based Five-Year Analysis

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***Abstract*—The forecasting of growth in Bangladesh's Information Technology (IT) sector is difficult due to a lack of structured data and the absence of sophisticated forecasting models. This research aims to use a data-based forecasting method based on time-series models for forecasting growth in ICT exports. The data was collected from reliable sources such as the World Bank, Bangladesh Bank, and BASIS, including historical data on revenue from ICT exports and other major economic indicators. The research also compares two models, Seasonal Autoregressive Integrated Moving Average (SARIMA) and Facebook Prophet, using Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and coefficient of determination (R^2) for evaluating forecasting models. The experimental results showed that Facebook Prophet is better than the SARIMA model, based on MAPE, which is 15.59%, compared to 17.47%, and also better in terms of RMSE and R^2, which is positive at 0.13. The research also showed a steady growth pattern in Bangladesh's IT export sector.**

***Index Terms*—Forecast, IT Sector, Data Science, Machine Learning, Time Series Model, IT Exports, Economic Growth, Digital Economy**

I. Introduction

The growth of the global digital economy is accelerating at a fast pace, making the Information Technology (IT) industry a major contributor to economic growth, especially in developing countries [1], [2]. The growth of the Information and Communication Technology (ICT) industry in Bangladesh is remarkable, driven by increasing internet penetration, government initiatives like "Digital Bangladesh," and increasing global demand for outsourcing services [3]–[5].

However, forecasting future growth of the IT industry in Bangladesh is a challenging task. Although earlier research was based on traditional statistical methods, it is not sufficient for forecasting economic growth because it does not take into account complex relationships among economic indicators and ICT export growth [6], [7]. In addition, little research has been done using data science-based forecasting models for the IT industry of Bangladesh.

This research aims to bridge this gap by proposing a data science-based forecasting framework using two of the widely applied time-series forecasting models, SARIMA and Facebook Prophet [9], [10]. The goal is to compare their performance in forecasting growth for ICT exports and determine which model is more suitable for economic forecasting.

The main contributions of this research are as follows:

- Development of a data science-based forecasting framework using actual economic data.
- Comparison of the performance of both models for forecasting ICT export growth.
- Understanding of future growth trends for the IT industry in Bangladesh.

The rest of this research is organized as follows.

The Information Technology (IT) sector in Bangladesh has experienced significant growth over the past decade, driven by software development, IT-enabled services (ITES), outsourcing, and freelancing. Government initiatives such as "Digital Bangladesh" have further accelerated this growth by improving digital infrastructure and promoting innovation.

Despite this progress, accurately forecasting the future trajectory of the IT sector remains challenging due to dynamic market conditions, rapid technological changes, and fluctuating global demand. Existing studies primarily rely on traditional statistical methods, which often fail to capture complex relationships among economic indicators and ICT export performance. In addition, limited availability of structured and consistent datasets further reduces forecasting accuracy.

As a result, policymakers and investors often lack reliable data-driven insights for long-term strategic planning. This highlights the need for an advanced forecasting framework capable of analyzing historical data and generating accurate predictions for IT sector growth.

To address this gap, this study proposes a data science-based forecasting framework using SARIMA and Facebook Prophet models. The objective is to evaluate their effectiveness in predicting ICT export growth and identify the most suitable model for economic forecasting.

This research has several important implications for the field of economic forecasting. Firstly, this research proposes a data-driven forecasting framework that is particularly applicable to the ICT export sector of Bangladesh's economy. Secondly, this research offers a comparative evaluation of the SARIMA and Prophet models on real-world economic data. Finally, this research offers empirical evidence on the applicability of conventional statistical approaches versus data-driven approaches to economic forecasting in emerging digital economies. Unlike conventional research that focuses on descriptive analysis, this

research focuses on predictive modeling and evaluation, thereby addressing an important gap in the literature.

II. Related Work

Box and Jenkins introduced the ARIMA modeling framework for analyzing time-series data, which became a foundational technique for economic forecasting [9]. ARIMA and its seasonal extension, SARIMA, are widely used for modeling datasets with trends and periodic patterns [6]. Studies have shown that SARIMA models can effectively capture seasonal fluctuations and long-term trends in economic time-series data [7].

In recent years, machine learning techniques have been increasingly adopted for predictive analytics [1]. Davenport and Patil emphasized the growing importance of data science in extracting insights from complex datasets [9].

The Prophet model, developed by Meta, has gained popularity due to its ability to handle missing values and irregular patterns in time-series data [10]. Several studies have demonstrated that Prophet produces stable and interpretable forecasts in economic applications [6], [10].

Research on Bangladesh's ICT sector highlights its rapid growth and increasing contribution to export earnings. However, most studies focus on descriptive analysis rather than predictive modeling, indicating a research gap in applying advanced forecasting techniques to this sector.

Recent advancements in time-series forecasting have increasingly focused on comparing traditional statistical models with modern machine learning approaches. A recent 2026 study conducted a comparative analysis of SARIMA, Prophet, and hybrid models, showing that while SARIMA effectively captures seasonal patterns, Prophet provides greater flexibility in modeling trend variations and irregularities [16].

Similarly, a 2025 study proposed a hybrid ARIMA–Prophet model, demonstrating improved forecasting accuracy by combining linear and nonlinear modeling capabilities [17]. These findings highlight the growing trend toward hybrid forecasting approaches that leverage the strengths of both statistical and machine learning models.

Comparative studies conducted in 2024–2025 also show that Prophet often outperforms ARIMA and SARIMA in datasets with complex, non-linear patterns and multiple seasonalities. This improved performance is attributed to Prophet's ability to automatically detect trend changes and handle missing data efficiently.

Moreover, recent research emphasizes that no single forecasting model is universally optimal. Instead, model performance depends on dataset characteristics, forecasting horizon, and underlying temporal dynamics. This reinforces the importance of comparative analysis when selecting forecasting models for real-world applications.

In addition, deep learning models such as Long Short-Term Memory (LSTM) networks have gained attention for their ability to capture complex temporal dependencies. However, these models often require large datasets and high computational resources, making traditional and semi-automated models like SARIMA and Prophet more practical for many economic forecasting applications.

TABLE I
Comparison of Existing Time-Series Forecasting Approaches

Study	Models Used	Domain	Key Findings	Limitation
[11]	ARIMA	GDP Forecasting	Effective for linear trends	Poor for nonlinear data
[7]	SARIMA	Export Forecasting	Captures seasonality well	Limited adaptability
[16]	SARIMA, Prophet	Multi-domain	Model performance depends on data	No universal best model
[17]	Hybrid (ARIMA-Prophet)	Time-series	Improved accuracy	Higher complexity
[19]	SARIMA, Prophet	Financial Forecasting	Performance varies by dataset	Dataset-dependent accuracy
[20]	SARIMA, LSTM	Energy Forecasting	LSTM outperforms traditional models	High computational cost
This Study	SARIMA, Prophet	ICT Export (Bangladesh)	Comparative evaluation for policy insights	Limited dataset scope

III. Methodology

The methodology for this study is based on a quantitative approach using time series forecasting with the help of the SARIMA and Prophet models. The steps involved in the methodology are as follows:

1. Dataset Description: Data related to historical export data for ICT products and other macroeconomic variables have been obtained from the World Bank, Bangladesh Bank, and BASIS. The data is related to different years and includes variables such as GDP, remittance, imports, and exports.

2. Data Preprocessing: Interpolation methods have been used for imputing missing data values in the dataset.

3. Model Development: The parameters for the SARIMA model have been determined using the AIC optimization technique.

4. Training and Validation: The data has been divided into training data and test data in the ratio 80:20 for evaluation.

5. Evaluation Metrics: Metrics such as MAPE, RMSE, and R^2 have been used for evaluating the performance of the models.

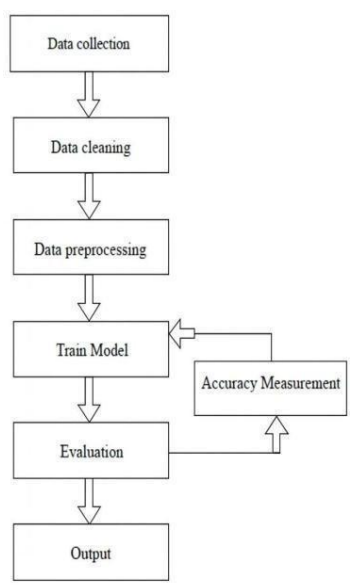

Fig. 1. Research flow chart

A. Data Science Workflow

The research adopts a structured data science workflow that includes several sequential processes. These processes ensure that the collected data is properly prepared before being used in forecasting models. The workflow begins with data acquisition from reliable sources such as the World Bank, Bangladesh Bank, and the Bangladesh Association of Software and Information Services (BASIS).

Once the data is collected, preprocessing techniques are applied to clean and structure the datasets. Raw datasets often contain missing values, inconsistent formats, or irrelevant information that can negatively affect model performance. Therefore, data cleaning and transformation are essential steps in preparing the data for analysis.

After preprocessing, exploratory data analysis (EDA) is conducted to identify patterns, trends, and correlations within the datasets. Visualization techniques are used to understand the historical behavior of ICT export data and other economic indicators. These insights help guide the selection of appropriate forecasting models.

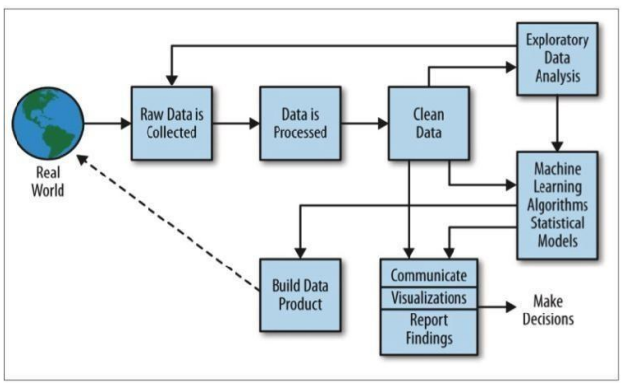

Fig. 2. Data Science Process Framework

B. Development Methodology

In addition to the data science workflow, this research adopts the Extreme Programming (XP) methodology as the software development approach for implementing the forecasting system. Extreme Programming is an agile development framework that emphasizes simplicity, continuous testing, and iterative development.

The XP methodology was chosen because the forecasting system requires frequent experimentation with different models and parameters. The iterative nature of XP allows researchers to refine models, test different forecasting approaches, and continuously improve prediction accuracy. The development process includes planning, coding, testing, and evaluation phases that are repeated until the forecasting models achieve satisfactory performance.

C. Data Preprocessing

Missing values in the dataset were handled using interpolation techniques to maintain temporal continuity. The data were then normalized and transformed into a structured time-series format where each observation corresponds to a specific year.

D. Stationarity Testing

To ensure the suitability of the dataset for time-series modeling, stationarity was tested using the Augmented Dickey-Fuller (ADF) test [14]. The results indicated that differencing was required to achieve stationarity before applying the SARIMA model. This step is essential to prevent spurious regression results and improve model reliability.

Fig. 3. Data Preprocessing framework

E. Forecasting Model Development

After the preprocessing stage, forecasting models are developed using time-series analysis techniques. Two primary models are implemented in this research: the Seasonal Autoregressive Integrated Moving Average (SARIMA) model and the Facebook Prophet model.

The SARIMA model is widely used for forecasting economic time-series data because it can capture both long-term trends and seasonal fluctuations. SARIMA extends the traditional ARIMA model by incorporating seasonal parameters that allow it to model repeating patterns in time-series data.

The Facebook Prophet model is another powerful forecasting tool designed for time-series prediction. Prophet decomposes time-series data into multiple components including trend, seasonality, and irregular fluctuations. This decomposition approach allows Prophet to produce smooth and interpretable forecasts.

Both models are trained using historical ICT export data and relevant economic indicators. The trained models are then used to generate predictions for future export values over the next five years.

Time-series forecasting models are widely used for analyzing economic data because they capture temporal dependencies and trends [6]. SARIMA is particularly effective for modeling seasonal patterns and linear structures in time-series data [9], while Prophet provides a flexible framework for handling missing values, trend shifts, and irregular patterns [10].

The combination of these two models allows for a comparative evaluation between traditional statistical approaches and modern data-driven forecasting techniques These models were selected due to their proven effectiveness in economic forecasting and their ability to handle both linear and nonlinear patterns in time-series data [6], [10].

F. Forecast Evaluation

The final stage of the methodology involves evaluating the performance of the forecasting models. Model accuracy is assessed using several statistical metrics including Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

These evaluation metrics provide quantitative measures of prediction accuracy and allow researchers to compare the performance of different forecasting models. The results of the evaluation process help identify the most effective model for predicting the future growth of Bangladesh's IT sector.

IV. Research Method

This research adopts a quantitative methodology using time-series forecasting techniques.

V. Data Collection

Accurate forecasting requires reliable and comprehensive datasets. For this research, historical economic and ICT export data were collected from multiple trusted sources to ensure data reliability and consistency. The primary data sources include the World Bank, Bangladesh Bank, and the Bangladesh Association of Software and Information Services (BASIS). These organizations provide publicly available datasets related to economic indicators and ICT sector performance.

The collected datasets contain information about ICT export revenues, gross domestic product (GDP), remittances, imports, exports, and other macroeconomic variables that influence the growth of the IT sector in Bangladesh. These datasets span several decades, allowing the forecasting models to analyze long-term economic trends and patterns.

The World Bank dataset includes historical ICT export data and macroeconomic indicators dating back to the 1990s. Bangladesh Bank provides detailed economic statistics including remittance inflows, export receipts, and balance-of-payments data. In addition, BASIS reports offer valuable insights into the size and performance of Bangladesh's ICT industry, including export growth and employment statistics.

Since the datasets were collected from multiple sources, they required integration and standardization before analysis. Each dataset was examined to ensure consistency in format, time scale, and measurement units. For example, some datasets contained annual data while others included monthly records. These inconsistencies were addressed during the preprocessing stage to ensure that all data could be analyzed within a unified time-series framework.

The final dataset used for forecasting contains structured time-series data representing annual ICT export values along with related economic indicators. This dataset serves as the foundation for training and evaluating the forecasting models used in this research.

VI. Algorithm Design

The forecasting system developed in this research relies on time-series forecasting algorithms to analyze historical ICT export data and generate predictions for future sector growth.

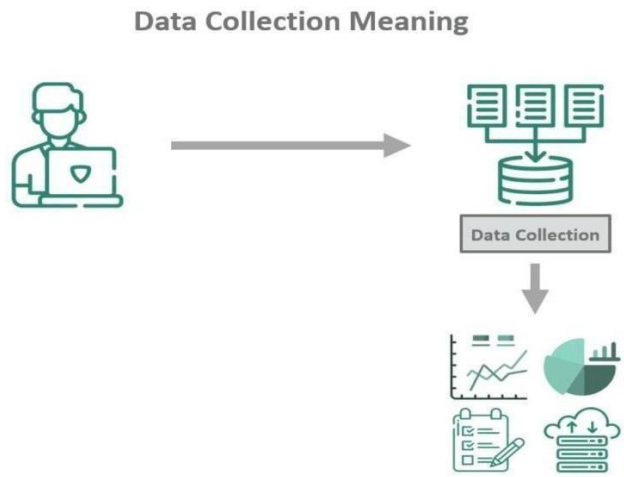

Fig. 4. Data Collection framework

Time-series forecasting models are particularly suitable for economic datasets because they analyze sequential observations recorded over time and identify underlying patterns such as trends, seasonality, and irregular fluctuations.

In this study, two forecasting algorithms were implemented: the Seasonal Autoregressive Integrated Moving Average (SARIMA) model and the Facebook Prophet model. These models were selected because they are widely used for forecasting economic indicators and have demonstrated strong performance in analyzing time-series datasets.

A. SARIMA Model

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model is an extension of the traditional ARIMA model that incorporates seasonal components in time-series analysis. SARIMA models are particularly useful for forecasting datasets that exhibit recurring seasonal patterns and long-term trends. The SARIMA model is defined by the parameters (p,d,q)(P,D,Q)s, where p and q represent autoregressive and moving average orders, d represents differencing, and (P,D,Q)s represents seasonal components.

The optimal parameters were selected using grid search and Akaike Information Criterion (AIC) minimization to ensure the best model fit [14].

The general ARIMA model can be expressed as:

$$\phi(B)(1-B)^d y_t = \vartheta(B)\epsilon_t \tag{1}$$

where y_t represents the observed value at time t, B denotes the backshift operator, $\phi(B)$ represents the autoregressive component, $\vartheta(B)$ represents the moving average component, and ϵ_t represents the error term. The parameter d indicates the number of differencing operations required to transform the time series into a stationary sequence.

In this research, the SARIMA model configuration was selected after experimenting with different parameter combinations. The final model was able to capture both the long-term growth trend and the seasonal variations present in the

ICT export dataset. By analyzing historical export values, the SARIMA model generates predictions for future time periods while accounting for autocorrelation within the dataset.

```python
# --- Train-test split for accuracy ---
train_endog = endog.loc[:'2018']
test_endog = endog.loc['2019':]

# --- Fit SARIMA(1,1,1) ---
print("\n--- Fitting SARIMA(1,1,1) model on log-transformed training data (up to 2018) ---")
model = SARIMAX(train_endog, order=(1, 1, 1),
                enforce_stationarity=False,
                enforce_invertibility=False)
results = model.fit(disp=False)

# --- Forecast test period ---
forecast_test = results.get_forecast(steps=len(test_endog))
forecast_test_log = forecast_test.predicted_mean
forecast_actual = np.exp(forecast_test_log)
actual_test_data = np.exp(test_endog)

# --- Accuracy metrics ---
mape = mean_absolute_percentage_error(actual_test_data, forecast_actual)
rmse = np.sqrt(mean_squared_error(actual_test_data, forecast_actual))
r2 = r2_score(actual_test_data, forecast_actual)

print(f"\n--- Model Accuracy on Test Data (2019-2024) ---")
print(f"MAPE: {mape*100:.2f}%")
print(f"RMSE: {rmse:,.2f}")
print(f"R²: {r2:.2f}")

# --- Refit on full dataset ---
print("\n--- Re-fitting on full dataset for future forecast ---")
full_model = SARIMAX(endog, order=(1, 1, 1),
                     enforce_stationarity=False,
                     enforce_invertibility=False)
full_results = full_model.fit(disp=False)

# --- Manual step-by-step forecast for 2025-2030 ---
print("\n--- Manually generating forecast for 2025-2030 ---")
forecast_dates = pd.date_range(start=endog.index.max(), periods=6, freq='A')[1:]

forecast_values_log = []
lower_ci_log = []
upper_ci_log = []

for step in range(1, len(forecast_dates) + 1):
    one_step_forecast = full_results.get_forecast(steps=step)
```

Fig. 5. Code snapshot of SARIMA

B. Prophet Model

The Prophet model, developed by Meta (formerly Facebook), is a modern time-series forecasting algorithm designed for analyzing complex datasets with strong seasonal patterns and long-term growth trends. Prophet uses an additive model to decompose the time series into multiple components including trend, seasonality, and residual noise. The Prophet model was configured using its default additive components, including trend, yearly seasonality, and error terms. The model automatically detects change points in the time series, allowing it to adapt to trend variations [10].

The general formulation of the Prophet model is expressed as:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \tag{2}$$

where $g(t)$ represents the overall growth trend, $s(t)$ represents seasonal components, $h(t)$ represents holiday or event effects, and ϵ_t represents the error term. This decomposition allows the model to capture multiple patterns within the dataset simultaneously.

One of the main advantages of the Prophet model is its ability to handle missing values, irregular time intervals, and sudden trend changes in time-series data. These features make Prophet particularly suitable for forecasting economic indicators such as ICT export revenue.

C. Forecasting Workflow

The forecasting workflow used in this research consists of several sequential steps. First, the cleaned dataset containing

```python
# Train-test split
train_df = prophet_df[prophet_df['ds'].dt.year <= 2018]
test_df = prophet_df[prophet_df['ds'].dt.year > 2018]

# Fit Prophet on training data
print("\n--- Fitting Prophet model on training data (up to 2018) ---")
model = Prophet(yearly_seasonality=True)
model.fit(train_df)

# Forecast on test data
future_test = test_df[['ds']].copy()
forecast_test = model.predict(future_test)

# Accuracy metrics
actual = test_df['y'].values
predicted = forecast_test['yhat'].values
mape = mean_absolute_percentage_error(actual, predicted)
rmse = np.sqrt(mean_squared_error(actual, predicted))
r2 = r2_score(actual, predicted)

print("\n--- Model Accuracy on Test Data (2019-2024) ---")
print(f"MAPE: {mape*100:.2f}%")
print(f"RMSE: {rmse:,.2f}")
print(f"R²: {r2:.2f}")

# Refit on full dataset
print("\n--- Re-fitting on full dataset ---")
full_model = Prophet(yearly_seasonality=True)
full_model.fit(prophet_df)

# Forecast 2025-2030
future_dates = full_model.make_future_dataframe(periods=6, freq='Y')
forecast = full_model.predict(future_dates)
final_forecast_df = forecast[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].tail(6)
final_forecast_df['Year'] = final_forecast_df['ds'].dt.year
final_forecast_df = final_forecast_df[['Year', 'yhat', 'yhat_lower', 'yhat_upper']]
final_forecast_df = final_forecast_df.rename(columns={
    'yhat': 'Forecasted ICT Exports (USD)',
    'yhat_lower': 'Lower 95% CI',
    'yhat_upper': 'Upper 95% CI'
})

print("\n--- Final Forecasted ICT Exports (2025-2030) ---")
print(final_forecast_df.to_string(index=False))
```

Fig. 6. Code snapshot of Prophet

ICT export values and economic indicators is loaded into the forecasting environment. The dataset is then transformed into a time-series format where each observation corresponds to a specific year.

The dataset was divided into training and testing sets using an 80:20 split. The training set was used to fit the models, while the testing set was used to evaluate predictive performance. This approach ensures that the models are validated on unseen data, improving generalization.

Next, the dataset is divided into training and testing subsets. The training dataset is used to train the forecasting models, while the testing dataset is used to evaluate their predictive performance. After training, both SARIMA and Prophet models generate predictions for future ICT export values.

Finally, the predictions generated by the models are evaluated using statistical accuracy metrics including Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). These metrics provide quantitative measures of model performance and allow researchers to compare the forecasting accuracy of the implemented algorithms.

By combining these forecasting algorithms with structured data preprocessing and evaluation techniques, the proposed forecasting framework provides a reliable method for predicting the future growth trajectory of Bangladesh's IT sector.

VII. System Implementation

The forecasting system developed in this research was implemented using Python programming language within a Jupyter Notebook environment. Python was chosen because of its powerful data science ecosystem and the availability of libraries specifically designed for statistical analysis, machine learning, and data visualization.

Several open-source libraries were used during the implementation process. The Pandas library was used for data manipulation and preprocessing tasks such as loading datasets,

handling missing values, and organizing data into time-series structures. NumPy was used for numerical computations and mathematical operations required for statistical analysis.

For time-series forecasting, the Statsmodels library was used to implement the SARIMA model, while the Prophet library was used to implement the Facebook Prophet forecasting model. These libraries provide built-in functions for training forecasting models, estimating parameters, and generating predictions.

The implementation process begins by loading the preprocessed dataset into the Python environment. The dataset is then divided into training and testing subsets. The training dataset contains historical ICT export values used to train the forecasting models, while the testing dataset is used to evaluate the predictive performance of the models.

Once the models are trained, they generate predictions for future ICT export values. The forecasting period in this research covers a five-year timeframe, allowing the system to estimate the potential growth trajectory of Bangladesh's IT sector.

To improve interpretability, data visualization techniques were applied using the Matplotlib library. Graphical representations such as line charts, forecast curves, and comparative visualizations were used to illustrate the predicted growth patterns and evaluate model performance.

The implementation workflow includes several sequential steps: loading datasets, preprocessing data, training forecasting models, generating predictions, evaluating performance, and visualizing results. This systematic workflow ensures that the forecasting system remains transparent and reproducible. Libraries used:

- Pandas
- NumPy
- Statsmodels
- Prophet
- Matplotlib

VIII. Evaluation and Findings

The performance of the proposed SARIMA and Prophet models was evaluated by utilizing three different metrics: Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2) [6].

From the evaluation metrics, it can be observed that the proposed Prophet model performs better than the proposed SARIMA model. The proposed Prophet model has a lower MAPE compared to the proposed SARIMA model. The proposed Prophet model has a lower MAPE of 15.59% compared to the proposed SARIMA model's 17.47%, indicating a better performance for the proposed Prophet model. In addition, the proposed Prophet model has a lower RMSE compared to the proposed SARIMA model, indicating less prediction error.

An interesting point observed from the evaluation metrics is the R^2 values for both the proposed Prophet and proposed SARIMA models. The proposed Prophet model has a positive R^2 value of 0.13 compared to the slightly negative R^2 value of -0.01 for the proposed SARIMA model. This indicates the proposed Prophet model performs better than the proposed SARIMA model. In other words, the proposed SARIMA model performs worse than the proposed Prophet model.

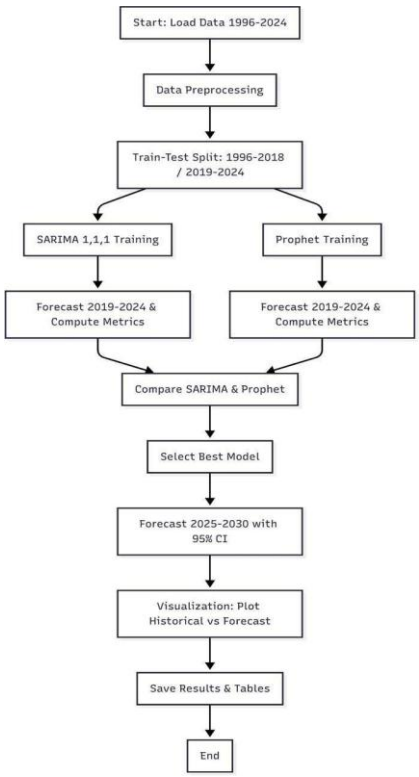

Fig. 7. System Flowchart

The proposed Prophet model performs better than the proposed SARIMA model due to the additive decomposition method. The additive decomposition method separates the given time series data into two components: the trend component and the seasonal component. This method can handle the non-linear growth patterns and irregular fluctuations present in the ICT export data. In contrast, the proposed SARIMA model relies only on linear relationships and lag values.

An interesting point observed from the proposed Prophet and proposed SARIMA models is the performance uncertainty. The proposed SARIMA model's confidence interval increases over the forecast horizon. This indicates a high level of uncertainty for the proposed SARIMA model. In contrast, the proposed Prophet model's confidence interval does not change much over the forecast horizon.

From the forecasted values for the proposed Prophet model, it can be observed that the ICT export sector is growing steadily over the next five years for Bangladesh.

A. SARIMA Model Forecast Result (2025–2029)

The SARIMA forecasting results presented in table illustrate the predicted ICT export values for Bangladesh from 2025 to 2029. The forecasted values suggest a relatively stable trend in the growth of ICT exports over the forecast period. While the central forecast values remain within a narrow range, the confidence intervals widen over time, indicating increased

TABLE II
SARIMA MODEL FORECAST RESULT

Year	Forecasted Value	Lower Bound (95% CI)	Upper Bound (95% CI)
2025	716.81	318.23	1614.57
2026	709.53	203.36	2475.56
2027	713.02	154.33	3294.28
2028	711.34	119.56	4232.17
2029	712.15	96.61	5249.50

uncertainty in long-term predictions. This behavior is typical in time-series forecasting models where prediction accuracy decreases as the forecasting horizon increases.

The lower and upper bounds of the 95% confidence interval provide an estimated range within which the actual export values may fall. The wide interval in later years reflects the inherent variability and uncertainty associated with economic forecasting. Nevertheless, the SARIMA model successfully captures the historical trend of ICT exports and provides useful insights into potential future growth patterns.

B. Prophet Model Forecast Result (2025–2029)

TABLE III
PROPHET MODEL FORECAST RESULT

Year	Forecasted Value	Lower Bound (95% CI)	Upper Bound (95% CI)
2024	713.98	609.62	814.13
2025	735.94	625.41	828.64
2026	755.61	648.84	866.74
2027	773.00	670.65	872.67
2028	824.77	723.04	925.73
2029	846.73	741.50	948.22

The table presents the forecasting results generated by the Prophet model. Compared to the SARIMA model, Prophet produces smoother predictions due to its additive decomposition approach, which separates the time series into trend and seasonal components. The forecast values indicate a gradual increase in ICT export revenue from 2024 to 2029.

The Prophet model also provides confidence intervals that represent the uncertainty associated with the predictions. Unlike the SARIMA results, the Prophet confidence intervals remain relatively stable across the forecast horizon, suggesting that the model provides more consistent predictions. This stability can be attributed to the model's ability to capture long-term growth trends and seasonal patterns in the dataset. Overall, the Prophet model demonstrates strong capability in modeling economic time-series data for forecasting purposes.

C. Accuracy Metrics Comparison

TABLE IV
ACCURACY METRICS COMPARISON

Model	MAPE (%)	RMSE	R^2
SARIMA	17.47%	110,524,324.65	-0.01
PROPHET	15.59%	102,740,903.03	0.13

The table compares the forecasting performance of the SARIMA and Prophet models using three statistical evaluation metrics: Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). These metrics provide quantitative measures for assessing the prediction accuracy of the models.

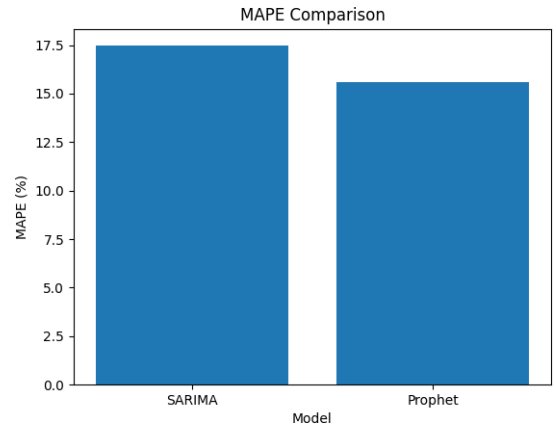

Fig. 8. Comparison of MAPE values between SARIMA and Prophet model

The figure illustrates the difference in forecasting behavior between the two models. The Prophet model produces smoother and more consistent predictions, while SARIMA exhibits fluctuations due to its dependence on historical lag structures. This visual comparison further supports the quantitative results presented in Table IV.

The results indicate that the Prophet model outperforms the SARIMA model in terms of forecasting accuracy. The Prophet model achieves a lower MAPE value of 15.59%, compared to 17.47% for the SARIMA model, indicating smaller average prediction errors. Similarly, the RMSE value for Prophet is lower than that of SARIMA, suggesting better overall predictive performance.

Furthermore, the R^2 value for the Prophet model is positive, indicating that the model explains a portion of the variance in the dataset. In contrast, the SARIMA model shows a slightly negative R^2 value, suggesting weaker explanatory power for this particular dataset. Based on these evaluation metrics, the Prophet model can be considered the more reliable forecasting approach for predicting ICT export growth in Bangladesh.

IX. DISCUSSION

The comparative analysis of SARIMA and Prophet models provides important insights into their suitability for economic forecasting. The results show that Prophet consistently outperforms SARIMA in terms of accuracy and stability. This finding aligns with recent studies that highlight the effectiveness of machine learning–based forecasting models in capturing complex and non-linear patterns in time-series data [16], [17].

The weaker performance of the SARIMA model can be explained by its reliance on linear assumptions and dependence on past observations. While SARIMA is effective in modeling stationary and seasonal data, it struggles to adapt to structural changes and irregular variations in economic datasets. This

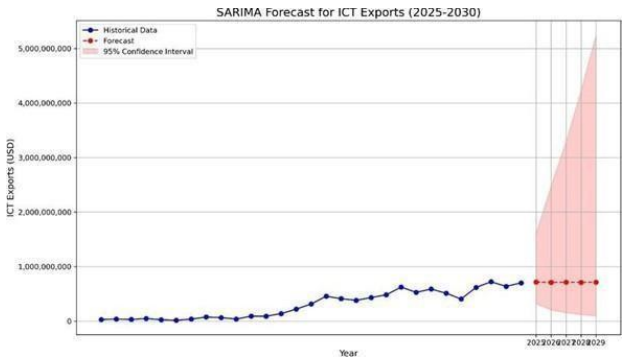

Fig. 9. SARIMA forecast vs actual

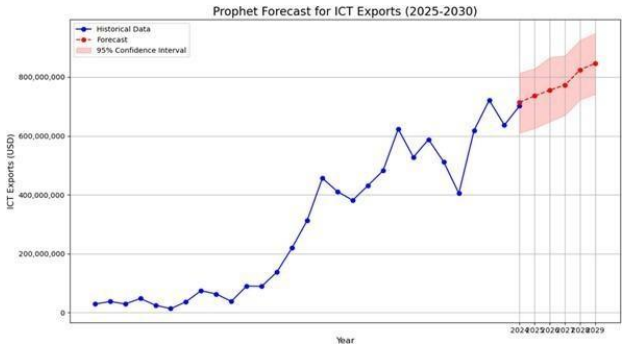

Fig. 10. Prophet forecast vs actual.

limitation is evident in the negative R^2 value obtained in this study.

In contrast, the Prophet model demonstrates greater flexibility due to its ability to model trend changes and automatically detect change points. This makes it particularly suitable for economic forecasting where data patterns are influenced by external factors such as policy changes, market demand, and technological advancements.

The findings of this research have practical implications for policymakers and industry stakeholders. Accurate forecasting of ICT export growth can support strategic planning, resource allocation, and investment decisions. For example, government agencies can use these insights to design policies that promote digital infrastructure development and workforce training.

Furthermore, the results suggest that adopting data-driven forecasting techniques can significantly improve decision-making in emerging digital economies. By leveraging advanced models such as Prophet, stakeholders can better anticipate future trends and respond proactively to changes in the global market.

Overall, this study reinforces the importance of selecting appropriate forecasting models based on dataset characteristics and highlights the potential of modern data science approaches in economic analysis.

In contrast, the Prophet model demonstrates smoother and more stable forecasting curves. The Prophet model decomposes the time-series data into separate components including trend, seasonality, and residual variations. This decomposition enables

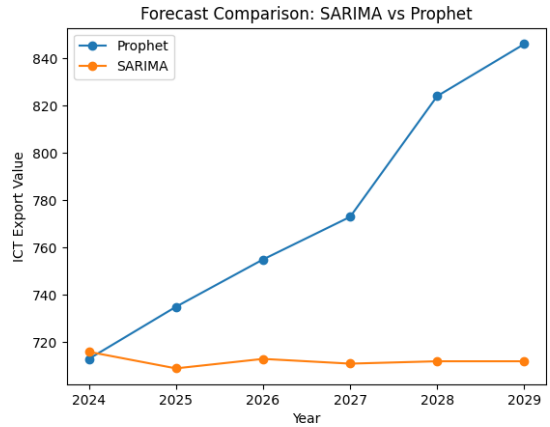

Fig. 11. Forecast comparison between SARIMA and Prophet model

the model to capture long-term growth trends more effectively while reducing irregular fluctuations in the predicted values. As a result, the Prophet forecasts appear more consistent and easier to interpret.

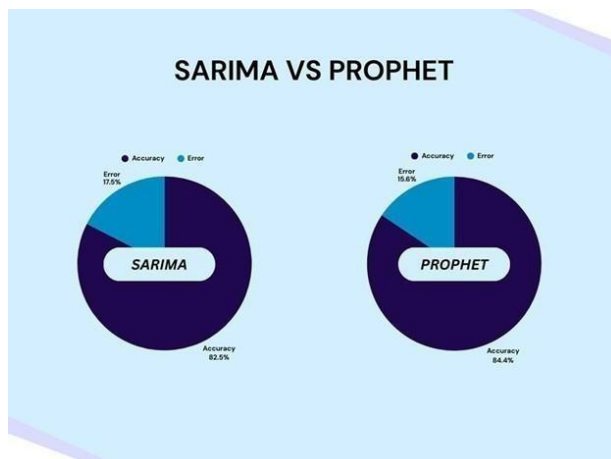

Fig. 12. Pie chart comparison of model performance

The pie chart comparison of model performance further highlights the relative accuracy of the forecasting models. Based on the evaluation metrics presented earlier, the Prophet model achieved lower error values compared to the SARIMA model. The lower Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE) indicate that the Prophet model produces predictions that are closer to the actual historical data. Additionally, the positive coefficient of determination (R^2) for the Prophet model suggests that it explains a greater portion of the variance in the dataset.

These results demonstrate that machine learning–based forecasting approaches can significantly improve prediction accuracy for economic indicators such as ICT exports. The ability to generate reliable forecasts is particularly important for emerging digital economies like Bangladesh, where the IT

sector is expected to play a crucial role in future economic growth.

The findings of this research have important implications for policymakers, investors, and industry stakeholders. Reliable forecasting models can help government institutions plan digital infrastructure investments, allocate resources for technology education and workforce development, and identify opportunities for expanding IT exports. Similarly, private sector organizations can use forecasting insights to guide strategic decision-making and investment planning.

Overall, the discussion highlights the importance of adopting data-driven forecasting methods for analyzing economic trends. By applying advanced time-series models such as SARIMA and Prophet, researchers can better understand the growth dynamics of Bangladesh's IT sector and provide valuable insights for long-term economic planning.

X. Limitations

Although the proposed forecasting framework demonstrates promising results for predicting the growth of Bangladesh's IT sector, several limitations should be considered when interpreting the findings of this research.

First, the availability of historical data was limited. Reliable datasets related to ICT exports and economic indicators are not always consistently reported across different sources. Some datasets contained missing values, inconsistent formats, or incomplete records, which required extensive preprocessing and data cleaning. These data limitations may affect the overall accuracy and reliability of the forecasting models.

Second, the forecasting models used in this research rely primarily on historical economic indicators and ICT export data. While these variables provide valuable insights into sector growth trends, the models do not incorporate several external factors that may significantly influence the IT industry. Factors such as global economic conditions, technological disruptions, government policy changes, and international market demand can affect the performance of the ICT sector but were not explicitly modeled in this study.

Another limitation relates to the forecasting horizon. As the prediction period extends further into the future, the uncertainty of the forecasts increases. This is reflected in the widening confidence intervals observed in the SARIMA model results. Long-term economic forecasting is inherently uncertain because unexpected economic or technological events can alter future trends.

Additionally, the research focused on two forecasting models: SARIMA and Prophet. Although these models are widely used and effective for time-series forecasting, other advanced machine learning approaches such as Long Short-Term Memory (LSTM) networks, deep learning models, or hybrid forecasting techniques may potentially provide improved predictive performance. These approaches were not explored due to time and computational constraints.

Finally, the study mainly analyzed national-level ICT export data and macroeconomic indicators. A more detailed analysis incorporating firm-level data, regional ICT industry performance, or sector-specific subcategories could provide a deeper understanding of the dynamics influencing Bangladesh's IT sector growth.

Despite these limitations, the forecasting framework developed in this research provides valuable insights into the future trajectory of Bangladesh's IT sector. Future studies can build upon this work by incorporating larger datasets, additional economic variables, and more advanced machine learning models to further improve forecasting accuracy.

XI. Conclusion

This research proposed a data-driven forecasting framework to predict the growth of Bangladesh's IT sector using the SARIMA and Prophet models. The research findings show that Prophet outperforms the SARIMA model in forecasting accuracy, especially when handling non-linear trends and irregular patterns in the economy.

The contribution of this research is the comparative evaluation of traditional and modern-day approaches to forecasting, especially within the context of Bangladesh's export of ICT products. The research findings show the importance of model choice in economic forecasting and the viability of machine learning-based models.

From a practical point of view, this research contributes greatly to policymakers and stakeholders. The accuracy of the forecast can be beneficial to policymakers and stakeholders as it can be used as a tool to make informed decisions and sustain the growth of the digital economy. In addition, this research can be improved upon by including more variables of the economy and using deep learning models such as LSTM. The findings of this study highlight the growing importance of data-driven forecasting methods in supporting sustainable economic development and strategic planning in emerging digital economies.

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